

CSE 713 — Artificial Intelligence

FOL, Resolution & Knowledge Representation

Complete Past Paper Solutions: 2020–2024

Source: *Propositional Logic.pdf* slides (all 104 pages read)

Every question from 2020–2024 included. Zero omissions.

Contents

1	2020 — Group-A Q4: Marcus/Pompeii (approx. 9 marks)	3
1.1	Part (a): WFF Translation	3
1.2	Part (b): Backward Reasoning — Is Marcus Alive?	3
1.3	Part (c): Clause Form Conversion	4
1.4	Part (d): Resolution Proof — Marcus is NOT Alive Now	4
2	2021 — Q4: PL Reasoning (approx. 9.25 marks)	6
2.1	Part (a): Wumpus World Knowledge Generation (3 marks)	6
2.2	Part (b): Proof vs. Theorem (0.75 marks)	7
2.3	Part (c): Satisfiability Analysis (2 marks)	7
2.4	Part (d): Resolution Proof — $P_{1,2}$ is False (3.5 marks)	8
3	2021 — Q5(a,b): FOL Generalization + Perfect Square (approx. 8.75 marks)	9
3.1	Part (a): FOL as Generalization of PL (1 mark)	9
3.2	Part (b-i): FOL Translation (1.75 marks)	9
3.3	Part (b-ii): Canonical Form / Clause Form (2.5 marks)	10
3.4	Part (b-iii): Resolution Proof with Unification (3.5 marks)	10
4	2022 — A-2: Resolution + Marcus/Pompeii (7 marks)	12
4.1	Part (a): Resolution + CFC Algorithm (2.5 marks)	12
4.2	Part (b): Marcus/Pompeii 2022 (“now = 2023”) (5 marks)	13
4.3	Part (c): Knowledge Representation and Mapping (1.5 marks)	13
5	2023 — Q4: FOL Translation + Drone (9 marks)	15
5.1	Part (a): Self-Driving Car FOL (5 marks)	15
5.2	Part (b): Drone Delivery FOL + Forward Chaining (4 marks)	16
6	2024 — Q4: ER + Perfect Square FOL (9 marks)	18
6.1	Part (a): Evidential Reasoning — Assessment of High-Level Attributes (1 mark)	18
6.2	Part (b): Combined Degree of Belief (1 mark)	18
6.3	Part (c-A): FOL Translation (2 marks)	19
6.4	Part (c-B): Proof Using Inference Rules (1.5 marks)	19
6.5	Part (c-C): Canonical Form Conversion (1.5 marks)	19
6.6	Part (c-D): Resolution Proof (2 marks)	20

Topic Overview (from Slides)

Topics Covered in This Document (from Propositional Logic.pdf)

Part 1 — Propositional Logic (PL):

- Modus Ponens & valid inference rules
- Satisfiability, Validity, Entailment
- Clausal form: literals, clauses (disjunctions of literals)
- 4-step conversion to clausal form in PL
- Resolution rule: $\frac{(x \vee s_1), (\neg x \vee s_2)}{s_1 \vee s_2}$
- Proof by Refutation (add negated goal $\rightarrow$ derive $\square$)

Part 2 — First Order Logic (FOL):

- Limitations of PL (cannot express infinite models)
- Terms: Constants, Variables, Functions; Predicates (return T/F)
- Quantifiers: $\forall$ (for all), $\exists$ (there exists), duality
- FOL sentences: combinations of predicates with $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$
- Inference rules: Universal Elimination (UE), Existential Elimination/Skolemization (EE), Existential Introduction (EI)
- Unification: finding substitution θ making two atoms identical
- 9-step Canonical Form Conversion (CFC) to clause form
- Resolution in FOL with Unification

9-Step Canonical Form Conversion Algorithm (exam-critical)

- Step 1.** Eliminate biconditionals: $A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$
- Step 2.** Eliminate implications: $A \rightarrow B \equiv \neg A \vee B$
- Step 3.** Move $\neg$ inward (De Morgan's, double-negation elimination)
- Step 4.** Standardize variables apart (rename so each clause has unique vars)
- Step 5.** Move all quantifiers to the left (prenex normal form)
- Step 6.** Eliminate existential quantifiers (**Skolemization**):
- $\exists y$ outside any $\forall$: replace y with a Skolem **constant** c
 - $\forall x \exists y$: replace y with a Skolem **function** $f(x)$
- Step 7.** Drop all $\forall$ (variables implicitly universally quantified)
- Step 8.** Distribute $\wedge$ over $\vee$ (convert to CNF)
- Step 9.** Write each conjunct as a separate clause (set of literals)

2020 — Group-A Q4: Marcus/Pompeii (approx. 9 marks)

Question — 2020 Group-A Q4

Consider the following facts:

- (i) Marcus was a man.
 - (ii) Marcus was a Pompeian.
 - (iii) Marcus was born in 40 A.D.
 - (iv) All men are mortal.
 - (v) All Pompeians died when the volcano erupted in 79 A.D.
 - (vi) No mortal lives longer than 150 years.
 - (vii) It is now 2021.
 - (viii) Alive means not dead.
 - (ix) If someone dies, then he is dead at all later times.
- (a) Translate these facts into well-formed formulas (WFFs) in predicate logic.
 - (b) Answer the question “Is Marcus alive now?” using backward reasoning.
 - (c) Convert the formulas into clause form.
 - (d) Prove that “Marcus is not alive now” using resolution.

Part (a): WFF Translation

WFF Translation — 2020 Q4(a)

- F_1 : $\text{man}(\text{Marcus})$
- F_2 : $\text{pompeian}(\text{Marcus})$
- F_3 : $\text{born}(\text{Marcus}, 40)$
- F_4 : $\forall x \text{ man}(x) \rightarrow \text{mortal}(x)$
- F_5 : $\forall x \text{ pompeian}(x) \rightarrow \text{dies}(x, 79)$
- F_6 : $\forall x \forall b \forall t \text{ mortal}(x) \wedge \text{born}(x, b) \wedge \text{gt}(t, \text{plus}(b, 150)) \rightarrow \text{dead}(x, t)$
- F_7 : $\text{now} = 2021$
- F_8 : $\forall x \text{ alive}(x) \leftrightarrow \neg \text{dead}(x, \text{now})$
- F_9 : $\forall x \forall t_1 \forall t_2 \text{ dies}(x, t_1) \wedge \text{gt}(t_2, t_1) \rightarrow \text{dead}(x, t_2)$

Part (b): Backward Reasoning — Is Marcus Alive?

Backward Reasoning — 2020 Q4(b)

Goal: Determine $\text{alive}(\text{Marcus})$. Start from the goal and work backwards through the rules.

Step 1. From F_8 : $\text{alive}(\text{Marcus}) \leftrightarrow \neg \text{dead}(\text{Marcus}, \text{now})$.

$\Rightarrow$ Need to determine whether $\text{dead}(\text{Marcus}, 2021)$ is true.

Step 2. To prove $\text{dead}(\text{Marcus}, 2021)$, use F_9 (with x/Marcus , $t_2/2021$):

Need: $\text{dies}(\text{Marcus}, t_1) \wedge \text{gt}(2021, t_1)$ for some t_1 .

Step 3. From F_5 (with x/Marcus): $\text{pompeian}(\text{Marcus}) \rightarrow \text{dies}(\text{Marcus}, 79)$.

From F_2 : $\text{pompeian}(\text{Marcus})$ is true. $\therefore \text{dies}(\text{Marcus}, 79)$. Set $t_1 = 79$.

Step 4. $\text{gt}(2021, 79) = \text{TRUE}$ (arithmetic fact).

Step 5. $\therefore \text{dead}(\text{Marcus}, 2021)$ (i.e., $\text{dead}(\text{Marcus}, \text{now})$) is TRUE.

Step 6. From F_8 : $\text{alive}(\text{Marcus}) \leftrightarrow \neg \text{dead}(\text{Marcus}, \text{now})$

Since $\text{dead}(\text{Marcus}, \text{now})$ is TRUE, $\text{alive}(\text{Marcus}) = \mathbf{FALSE}$.

Answer: Marcus is NOT alive now.

Part (c): Clause Form Conversion

Clause Form — 2020 Q4(c)

Apply the 9-step CFC algorithm:

Formula	Steps Applied	Clause(s)
F_1 : $\text{man}(\text{Marcus})$	ground atom	$C_1 : \{\text{man}(\text{Marcus})\}$
F_2 : $\text{pompeian}(\text{Marcus})$	ground atom	$C_2 : \{\text{pompeian}(\text{Marcus})\}$
F_3 : $\text{born}(\text{Marcus}, 40)$	ground atom	$C_3 : \{\text{born}(\text{Marcus}, 40)\}$
F_4 : $\forall x \text{man}(x) \rightarrow \text{mortal}(x)$	Steps 2,7	$C_4 : \{\neg \text{man}(x), \text{mortal}(x)\}$
F_5 : $\forall x \text{pompeian}(x) \rightarrow \text{dies}(x, 79)$	Steps 2,7	$C_5 : \{\neg \text{pompeian}(x), \text{dies}(x, 79)\}$
F_6 : $\text{mortal} + \text{born} + \text{gt} \rightarrow \text{dead}$	Steps 2,7	$C_6 : \{\neg \text{mortal}(x), \neg \text{born}(x, b), \neg \text{gt}(t, \text{plus}(b, 1))\}$
F_7 : $\text{now} = 2021$	fact	$C_7 : \{\text{now} = 2021\}$ (arithmetic)
F_8 : $\forall x \text{alive}(x) \leftrightarrow \neg \text{dead}(x, \text{now})$	Steps 1,2,7,9	$C_8 : \{\neg \text{alive}(x), \neg \text{dead}(x, \text{now})\}$ $C_9 : \{\text{dead}(x, \text{now}), \text{alive}(x)\}$
F_9 : $\text{dies} \wedge \text{gt} \rightarrow \text{dead}$	Steps 2,7	$C_{10} : \{\neg \text{dies}(x, t_1), \neg \text{gt}(t_2, t_1), \text{dead}(x, t_2)\}$

Part (d): Resolution Proof — Marcus is NOT Alive Now

Resolution Proof by Refutation — 2020 Q4(d)

Strategy: Negate the goal. Add $C_{\neg} : \{\text{alive}(\text{Marcus})\}$ and derive the empty clause $\square$.

Step	Clauses Used	Substitution	Result
1	C_1 & C_4	$\{x/\text{Marcus}\}$	$C_{11} : \{\text{mortal}(\text{Marcus})\}$
2	C_2 & C_5	$\{x/\text{Marcus}\}$	$C_{12} : \{\text{dies}(\text{Marcus}, 79)\}$
3	C_{12} & C_{10}	$\{x/\text{Marcus}, t_1/79\}$	$C_{13} : \{\neg \text{gt}(t_2, 79), \text{dead}(\text{Marcus}, t_2)\}$
4	$C_{13} + \text{gt}(2021, 79) \text{ true}$	$\{t_2/2021\}$ (now=2021)	$C_{14} : \{\text{dead}(\text{Marcus}, 2021)\}$ $\equiv \{\text{dead}(\text{Marcus}, \text{now})\}$
5	$C_{\neg}$ & C_8	$\{x/\text{Marcus}\}$	$C_{15} : \{\neg \text{dead}(\text{Marcus}, \text{now})\}$
6	C_{14} & C_{15}	—	$\square$ (Contradiction!)

Conclusion: Adding $\text{alive}(\text{Marcus})$ to the KB leads to contradiction.

∴ alive(Marcus) is FALSE, i.e., **Marcus is NOT alive now.** ✓

2021 — Q4: PL Reasoning (approx. 9.25 marks)

Question — 2021 Q4

- (a) The figure shows a Wumpus World game. The agent starts from [1,1]. Briefly discuss with diagrams how the agent generates new knowledge through perceiving the environment and choosing its next move. (3 marks)
- (b) Differentiate between proof and theorem in one point. (0.75 marks)
- (c) Determine whether the following sentence is (i) Satisfiable, (ii) Contradictory, or (iii) Valid:

$$(P \vee Q) \wedge (P \vee \neg Q) \vee P$$

(2 marks)

- (d) Given knowledge base R_1 – R_5 below, show that $P_{1,2}$ is false using resolution: (3.5 marks)

$$R_1 : \neg P_{1,1} \quad R_2 : B_{1,1} \leftrightarrow (P_{2,1} \vee P_{1,2}) \quad R_3 : B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4 : \neg B_{1,1} \quad R_5 : B_{2,1}$$

Part (a): Wumpus World Knowledge Generation (3 marks)

Wumpus World — 2021 Q4(a)

The Wumpus World uses **Propositional Logic (PL)** to reason about hidden dangers. The agent builds a KB incrementally from perceptions.

Key Propositions:

- $B_{x,y}$: Breeze is felt at cell $[x, y]$ $P_{x,y}$: Pit exists at $[x, y]$
- $S_{x,y}$: Stench is felt at $[x, y]$ $W_{x,y}$: Wumpus exists at $[x, y]$

Initial KB rules (KB axioms):

$$R_1 : \neg P_{1,1} \quad (\text{no pit at start})$$

$$R_2 : B_{1,1} \leftrightarrow (P_{2,1} \vee P_{1,2})$$

$$R_4 : \neg B_{1,1} \quad (\text{agent perceives no breeze at } [1,1])$$

Inference chain:

1. Agent at [1,1]: perceives $\neg B_{1,1}$ (no breeze), $\neg S_{1,1}$ (no stench).
2. From $\neg B_{1,1} + R_2$ (biconditional left-to-right negated):
 $\neg B_{1,1} \rightarrow \neg(P_{2,1} \vee P_{1,2}) \equiv \neg P_{2,1} \wedge \neg P_{1,2}$
 $\therefore$ Cells [2,1] and [1,2] are **PIT-FREE** (safe to enter).
3. Agent moves to [2,1]: perceives $B_{2,1}$ (breeze felt). From $R_5 : B_{2,1}$ is added to KB.
4. From $R_3 : B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 $B_{2,1}$ is TRUE $\Rightarrow$ at least one of $\{P_{1,1}, P_{2,2}, P_{3,1}\}$ is true.
 $\neg P_{1,1}$ (from R_1) is known. $\neg P_{2,2}$ can be inferred (no path explored).
 $\therefore P_{3,1}$ is TRUE — **Pit at [3,1]!**

5. Agent **avoids** [3,1], backtracks, and continues to [1,2] (safe from Step 2).

Each perception updates the KB; logical inference (resolution) derives new facts about safe/dangerous cells without direct observation.

Part (b): Proof vs. Theorem (0.75 marks)

Proof vs. Theorem — 2021 Q4(b)

- A **theorem** is a statement (proposition) that has been established as true within a formal system.
- A **proof** is the sequence of logical steps (inference rules, axioms) that *establishes* the truth of a theorem.

In one point: A theorem is the **conclusion**; a proof is the **derivation** that justifies it.

Part (c): Satisfiability Analysis (2 marks)

Satisfiability of $(P \vee Q) \wedge (P \vee \neg Q) \vee P$ — 2021 Q4(c)

Step 1: Apply operator precedence ($\neg > \wedge > \vee$):

$$(P \vee Q) \wedge (P \vee \neg Q) \vee P = [(P \vee Q) \wedge (P \vee \neg Q)] \vee P$$

Step 2: Simplify the bracketed expression using Distributivity:

$$(P \vee Q) \wedge (P \vee \neg Q) = P \vee (Q \wedge \neg Q) = P \vee F = P$$

Step 3: Final expression:

$$P \vee P = P$$

Step 4: Truth table check:

P	Expression value
T	T
F	F

Conclusion:

- **Satisfiable** — TRUE when $P = T$. ✓
- NOT Contradictory (not always false)
- NOT Valid (not always true; false when $P = F$)

Part (d): Resolution Proof — $P_{1,2}$ is False (3.5 marks)**KB Resolution — Show $P_{1,2}$ is FALSE — 2021 Q4(d)****Given KB:**

$$R_1 : \neg P_{1,1}$$

$$R_2 : B_{1,1} \leftrightarrow (P_{2,1} \vee P_{1,2})$$

$$R_3 : B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4 : \neg B_{1,1}$$

$$R_5 : B_{2,1}$$

Convert R_2 to clause form (Steps 1–2 of CFC):

$$R_2 \equiv \underbrace{(B_{1,1} \rightarrow P_{2,1} \vee P_{1,2})}_{C_a: \neg B_{1,1} \vee P_{2,1} \vee P_{1,2}} \wedge \underbrace{(P_{2,1} \vee P_{1,2} \rightarrow B_{1,1})}_{C_b: \neg P_{2,1} \vee \neg B_{1,1} \text{ and } C_c: \neg P_{1,2} \vee \neg B_{1,1}}$$

Goal: Prove $\neg P_{1,2}$ (i.e., $P_{1,2}$ is false).**Method:** Refutation — add $\{P_{1,2}\}$ (negation of goal) to KB, derive $\square$.

Step	Clauses Resolved	Result
1	$\{P_{1,2}\} + C_c : \{\neg P_{1,2} \vee B_{1,1}\}$	$\{B_{1,1}\}$
2	$\{B_{1,1}\} + R_4 : \{\neg B_{1,1}\}$	$\square$ (Contradiction!)

Since assuming $P_{1,2}$ leads to contradiction, $P_{1,2}$ is **FALSE**. ✓**Intuition:** R_4 says no breeze at [1,1]. R_2 then tells us no pit at [2,1] or [1,2].

2021 — Q5(a,b): FOL Generalization + Perfect Square (approx. 8.75 marks)

Question — 2021 Q5(a) & Q5(b)

- (a) Briefly explain why FOL is considered as the generalization of PL. (1 mark)
- (b) Consider the following sentences: (7.75 marks)
- If a perfect square is divisible by a prime P , then it is also divisible by the square of P .
 - Every perfect square is divisible by some prime.
 - 36 is a perfect square.
 - Does there exist a prime q such that the square of q divides 36?

Prove via the resolution method and unification (where needed) that “Does there exist a prime q such that the square of q divides 36?” More precisely:

- First, translate the sentences above into FOL sentences.
- Convert to Clause Forms.
- Apply the resolution method with unification (where needed) to prove the goal sentence.

Part (a): FOL as Generalization of PL (1 mark)

FOL vs. PL — 2021 Q5(a)

PL can only express ground propositions (true/false facts about specific named objects). It **cannot** handle statements about all members of a set without listing them individually.

FOL extends PL by adding:

- Variables** ranging over objects (not just truth values)
- Universal quantifier** $\forall$: “for all objects of a given type”
- Existential quantifier** $\exists$: “there exists at least one object”
- Predicates** (relations on objects) and **Functions** (maps objects to objects)

Example: “All men are mortal” requires infinitely many PL sentences (mortal(Marcus), mortal(Brutus), ...) but only ONE FOL sentence: $\forall x \text{ man}(x) \rightarrow \text{mortal}(x)$.

Part (b-i): FOL Translation (1.75 marks)

FOL Translation — 2021 Q5(b-i)

Predicates declared:

- $\text{PerfSq}(x)$: x is a perfect square
- $\text{Prime}(p)$: p is a prime number
- $\text{Divides}(a, x)$: a divides x
- $\text{square}(p)$: function returning p^2

$$F_1 : \quad \forall x \forall p \text{ PerfSq}(x) \wedge \text{Prime}(p) \wedge \text{Divides}(p, x) \rightarrow \text{Divides}(\text{square}(p), x) \quad (1)$$

$$F_2 : \quad \forall x \text{ PerfSq}(x) \rightarrow \exists p \text{ Prime}(p) \wedge \text{Divides}(p, x) \quad (2)$$

$$F_3 : \quad \text{PerfSq}(36) \quad (3)$$

$$\text{Goal} : \quad \exists q \text{ Prime}(q) \wedge \text{Divides}(\text{square}(q), 36) \quad (4)$$

Part (b-ii): Canonical Form / Clause Form (2.5 marks)

Clause Form Conversion — 2021 Q5(b-ii)

Clause from F_1 (Steps 2, 7):

$$\begin{aligned} F_1 &\rightarrow \neg \text{PerfSq}(x) \vee \neg \text{Prime}(p) \vee \neg \text{Divides}(p, x) \vee \text{Divides}(\text{square}(p), x) \\ &\Rightarrow C_1 : \{ \neg \text{PerfSq}(x), \neg \text{Prime}(p), \neg \text{Divides}(p, x), \text{Divides}(\text{square}(p), x) \} \end{aligned}$$

Clause from F_2 (Steps 2, 6 Skolemization, 7):

$$F_2 : \forall x \text{ PerfSq}(x) \rightarrow \exists p \text{ Prime}(p) \wedge \text{Divides}(p, x)$$

The $\exists p$ is inside $\forall x$, so replace p with Skolem function $g(x)$:

$$\begin{aligned} &\forall x \text{ PerfSq}(x) \rightarrow \text{Prime}(g(x)) \wedge \text{Divides}(g(x), x) \\ \text{Split: } &C_{2a} : \{ \neg \text{PerfSq}(x), \text{Prime}(g(x)) \} \\ &C_{2b} : \{ \neg \text{PerfSq}(x), \text{Divides}(g(x), x) \} \end{aligned}$$

Clause from F_3 :

$$C_3 : \{ \text{PerfSq}(36) \}$$

Negated Goal (negate F_{goal} for refutation):

$$\neg \exists q \text{ Prime}(q) \wedge \text{Divides}(\text{square}(q), 36) \equiv \forall q \neg \text{Prime}(q) \vee \neg \text{Divides}(\text{square}(q), 36)$$

$$C_4 : \{ \neg \text{Prime}(q), \neg \text{Divides}(\text{square}(q), 36) \}$$

Part (b-iii): Resolution Proof with Unification (3.5 marks)

Resolution Proof — 2021 Q5(b-iii)

Clause set: C_1, C_{2a}, C_{2b}, C_3 (KB) + C_4 (negated goal)

Step	Resolved	Unifier	Result
1	$C_3 + C_{2a}$	$\{x/36\}$	$C_5 : \{\text{Prime}(g(36))\}$
2	$C_3 + C_{2b}$	$\{x/36\}$	$C_6 : \{\text{Divides}(g(36), 36)\}$
3	$C_3 + C_1$	$\{x/36\}$	$C_7 : \{\neg \text{Prime}(p), \neg \text{Divides}(p, 36), \text{Divides}(\text{square}(p), 36)\}$
4	$C_5 + C_7$	$\{p/g(36)\}$	$C_8 : \{\neg \text{Divides}(g(36), 36), \text{Divides}(\text{square}(g(36)), 36)\}$
5	$C_6 + C_8$	—	$C_9 : \{\text{Divides}(\text{square}(g(36)), 36)\}$
6	$C_5 + C_4$	$\{q/g(36)\}$	$C_{10} : \{\neg \text{Divides}(\text{square}(g(36)), 36)\}$
7	$C_9 + C_{10}$	—	$\square$ (Contradiction!)

Conclusion: Negating the goal leads to contradiction. Therefore:

$\exists q$ such that $\text{Prime}(q) \wedge \text{Divides}(q^2, 36)$ is **TRUE**. ✓

Note: $g(36)$ is the Skolem function instantiated at 36. It represents the specific prime (e.g., 2 or 3) that divides 36. In reality, $2^2 = 4$ divides 36, and $3^2 = 9$ divides 36.

2022 — A-2: Resolution + Marcus/Pompeii (7 marks)

Question — 2022 A-2

- (a) What is resolution? Write down the algorithm for the conversion of well-formed formulas (wff) in predicate logic to clause form. (2.5 marks)
- (b) Consider the following facts (F-1 to F-9) about Marcus and Pompeii. “It is now 2023.”
- (i) Translate these facts into wff in predicate logic. (1 mark)
 - (ii) Answer the question “Is Marcus alive now?” using backward reasoning. (1 mark)
 - (iii) Convert the formula into clause form. (1 mark)
 - (iv) Prove that “Marcus is not alive now” using resolution. (2 marks)
- (c) What do you mean by Knowledge Representation and Mapping? What are the roles of Knowledge Representation? (1.5 marks)

Part (a): Resolution + CFC Algorithm (2.5 marks)

Resolution + CFC Algorithm — 2022 A-2(a)

Resolution is a refutation-complete inference procedure that operates on clauses. Given two clauses containing complementary literals (L and $\neg L$), resolution derives a new clause (the *resolvent*) with those literals removed:

$$\frac{(L \vee \alpha_1 \vee \cdots \vee \alpha_n) \quad (\neg L \vee \beta_1 \vee \cdots \vee \beta_m)}{(\alpha_1 \vee \cdots \vee \alpha_n \vee \beta_1 \vee \cdots \vee \beta_m)}$$

In FOL, we use **unification** to find a substitution θ making $L\theta = L'\theta$ before resolving.

Algorithm for wff $\rightarrow$ Clause Form (9 steps):

- Step 1. Eliminate biconditionals:** $A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$
- Step 2. Eliminate implications:** $A \rightarrow B \equiv \neg A \vee B$
- Step 3. Move $\neg$ inward:** Apply De Morgan’s laws; eliminate double negations
- Step 4. Standardize variables:** Rename vars so each clause has distinct variable names
- Step 5. Prenex normal form:** Move all quantifiers to the left
- Step 6. Skolemization** (eliminate $\exists$):
- $\exists y$ outside all $\forall$: replace y with Skolem constant c
 - $\forall x_1 \cdots x_n \exists y$: replace y with Skolem function $f(x_1, \dots, x_n)$
- Step 7. Drop all $\forall$**
- Step 8. Distribute $\wedge$ over $\vee$** (convert to CNF)
- Step 9. Write as set of clauses** (each conjunct is a clause = set of literals)

Part (b): Marcus/Pompeii 2022 (“now = 2023”) (5 marks)**WFF Translation — 2022 A-2(b-i)**

Same as 2020 but with now = 2023:

$$F_1 : \text{man}(\text{Marcus}) \quad F_2 : \text{pompeian}(\text{Marcus}) \quad F_3 : \text{born}(\text{Marcus}, 40)$$

$$F_4 : \forall x \text{ man}(x) \rightarrow \text{mortal}(x)$$

$$F_5 : \forall x \text{ pompeian}(x) \rightarrow \text{dies}(x, 79)$$

$$F_6 : \forall x \forall b \forall t \text{ mortal}(x) \wedge \text{born}(x, b) \wedge \text{gt}(t, \text{plus}(b, 150)) \rightarrow \text{dead}(x, t)$$

$$F_7 : \text{now} = 2023$$

$$F_8 : \forall x \text{ alive}(x) \leftrightarrow \neg \text{dead}(x, \text{now})$$

$$F_9 : \forall x \forall t_1 \forall t_2 \text{ dies}(x, t_1) \wedge \text{gt}(t_2, t_1) \rightarrow \text{dead}(x, t_2)$$
Backward Reasoning — 2022 A-2(b-ii)

Identical chain to 2020, but now = 2023. Since $\text{gt}(2023, 79)$ is TRUE, the chain holds:

$$\text{pompeian}(\text{Marcus}) \xrightarrow{F_5} \text{dies}(\text{Marcus}, 79) \xrightarrow{F_9, \text{gt}(2023, 79)} \text{dead}(\text{Marcus}, 2023) \xrightarrow{F_8} \neg \text{alive}(\text{Marcus})$$

Answer: Marcus is NOT alive in 2023.

Clause Form — 2022 A-2(b-iii)

Clause set is identical to 2020 (Section 1, Part c), with now = 2023. Key clauses:

$$C_4 : \{\neg \text{man}(x), \text{mortal}(x)\}$$

$$C_5 : \{\neg \text{pompeian}(x), \text{dies}(x, 79)\}$$

$$C_8 : \{\neg \text{alive}(x), \neg \text{dead}(x, \text{now})\}$$

$$C_{10} : \{\neg \text{dies}(x, t_1), \neg \text{gt}(t_2, t_1), \text{dead}(x, t_2)\}$$
Resolution Proof — 2022 A-2(b-iv)

Same steps as 2020 Section 1 Part (d), with now=2023.

Refutation: Add $\{\text{alive}(\text{Marcus})\}$, derive $\square$ in 6 steps. (See Section 1 for full trace.)

Conclusion: Marcus is NOT alive now. ✓

Part (c): Knowledge Representation and Mapping (1.5 marks)**Knowledge Representation — 2022 A-2(c)**

Knowledge Representation (KR) is the field concerned with encoding world knowledge in a formal structure that AI systems can use to reason, solve problems, and make decisions.

Mapping refers to the correspondence between real-world entities/relationships and their formal symbols in the KR system. For example, the fact “Marcus is a man” maps to the atom $\text{man}(\text{Marcus})$.

Roles of Knowledge Representation:

1. **Surrogate:** Substitutes for real-world entities, enabling reasoning without physical interaction.
2. **Ontological commitment:** Defines what exists in the domain (the “things” the system knows about).
3. **Inferential foundation:** Supports derivation of new facts from existing ones via inference rules.
4. **Medium of expression:** A formal language for humans to communicate domain knowledge to machines.
5. **Efficient computation:** Structured representations (clauses, networks) support efficient reasoning algorithms.

2023 — Q4: FOL Translation + Drone (9 marks)

Question — 2023 Q4

- (a) Define the following actions of a self-driving car using First-Order Logic (FOL). Declare the predicates first. (5 marks)
- (i) The car can move to a destination if the destination is reachable and there are no obstacles on the road.
 - (ii) A self-driving car can proceed through an intersection if the traffic light is green and there are no pedestrians crossing.
 - (iii) How would you represent in FOL that a self-driving car should stop if there is an obstacle on the road?
 - (iv) A self-driving car should reduce its speed when entering a school zone if the school zone is active (e.g., during certain hours of the day).
 - (v) Represent in FOL the rule that a self-driving car can only change lanes if the adjacent lane is clear and the car is moving.
- (b) A drone is responsible for delivering a package to a destination. Drone starts at L_0 , air route R_1 connects L_0 to L_1 , R_2 connects L_1 to L_2 (**obstacle on R_2**), and R_3 connects L_2 to L_p (destination). Define the predicates, knowledge base, and rules in FOL, then use forward chaining to determine if the drone can reach the destination. (4 marks)

Part (a): Self-Driving Car FOL (5 marks)

Self-Driving Car FOL — 2023 Q4(a)

Predicate declarations:

- $\text{Car}(c)$: c is a self-driving car
- $\text{Reachable}(c, d)$: destination d is reachable by car c
- $\text{Obstacle}(d)$: there is an obstacle at/on road to d
- $\text{Moves}(c, d)$: car c moves to destination d
- $\text{AtIntersection}(c, i)$: car c is at intersection i
- $\text{GreenLight}(i)$: traffic light at i is green
- $\text{PedestrianCrossing}(i)$: pedestrian is crossing at i
- $\text{Proceeds}(c, i)$: car c proceeds through intersection i
- $\text{ObstacleOnRoad}(c)$: obstacle on road ahead of car c
- $\text{Stops}(c)$: car c stops
- $\text{InSchoolZone}(c, z)$: car c is in school zone z
- $\text{ActiveHours}(z)$: school zone z is currently active
- $\text{ReduceSpeed}(c)$: car c reduces speed
- $\text{AdjacentLaneClear}(c, l)$: lane l adjacent to car c 's lane is clear
- $\text{Moving}(c)$: car c is moving
- $\text{ChangeLane}(c, l)$: car c changes to lane l

- (i) The car can move to a destination if reachable and no obstacles:

$$\forall c \forall d \text{ Car}(c) \wedge \text{Reachable}(c, d) \wedge \neg \text{Obstacle}(d) \rightarrow \text{Moves}(c, d)$$

(ii) Proceed through intersection if green light and no pedestrians:

$$\forall c \forall i \text{ Car}(c) \wedge \text{AtIntersection}(c, i) \wedge \text{GreenLight}(i) \wedge \neg \text{PedestrianCrossing}(i) \rightarrow \text{Proceeds}(c, i)$$

(iii) Stop if obstacle on road:

$$\forall c \text{ Car}(c) \wedge \text{ObstacleOnRoad}(c) \rightarrow \text{Stops}(c)$$

(iv) Reduce speed in school zone during active hours:

$$\forall c \forall z \text{ Car}(c) \wedge \text{InSchoolZone}(c, z) \wedge \text{ActiveHours}(z) \rightarrow \text{ReduceSpeed}(c)$$

(v) Change lanes only if adjacent lane clear and car is moving:

$$\forall c \forall l \text{ Car}(c) \wedge \text{AdjacentLaneClear}(c, l) \wedge \text{Moving}(c) \rightarrow \text{ChangeLane}(c, l)$$

Part (b): Drone Delivery FOL + Forward Chaining (4 marks)

Drone FOL + Forward Chaining — 2023 Q4(b)

Predicate declarations:

- $\text{At}(d, l)$: drone d is at location l
- $\text{Route}(l_1, l_2)$: there exists an air route from l_1 to l_2
- $\text{ObstacleOnRoute}(l_1, l_2)$: there is an obstacle on the route from l_1 to l_2
- $\text{CanFly}(d, l_1, l_2)$: drone d can fly from l_1 to l_2

Knowledge Base (Facts):

$$\begin{aligned} KB_1 &: \text{At}(\text{drone}, L_0) \\ KB_2 &: \text{Route}(L_0, L_1) \quad (\text{route R1 — clear}) \\ KB_3 &: \text{Route}(L_1, L_2) \quad (\text{route R2 — has obstacle!}) \\ KB_4 &: \text{Route}(L_2, L_p) \quad (\text{route R3 — clear}) \\ KB_5 &: \text{ObstacleOnRoute}(L_1, L_2) \quad (\text{obstacle on R2}) \\ KB_6 &: \neg \text{ObstacleOnRoute}(L_0, L_1) \quad (\text{R1 is clear}) \\ KB_7 &: \neg \text{ObstacleOnRoute}(L_2, L_p) \quad (\text{R3 is clear}) \end{aligned}$$

Movement Rule in FOL:

$$\forall d \forall l_1 \forall l_2 \text{ At}(d, l_1) \wedge \text{Route}(l_1, l_2) \wedge \neg \text{ObstacleOnRoute}(l_1, l_2) \rightarrow \text{At}(d, l_2)$$

Forward Chaining (data-driven):

Iteration	Antecedents matched	Conclusion (added to KB)
1	$\text{At}(\text{drone}, L_0) + \text{Route}(L_0, L_1) + \neg \text{Obstacle}(L_0, L_1)$	$\text{At}(\text{drone}, L_1)$
2	$\text{At}(\text{drone}, L_1) + \text{Route}(L_1, L_2) + \text{ObstacleOnRoute}(L_1, L_2)$ Condition $\neg \text{ObstacleOnRoute}(L_1, L_2)$ is FALSE	RULE FAILS No inference

Conclusion: Drone reaches L_1 successfully, but **cannot proceed to L_2** due to the obstacle on route R_2 . Therefore, the drone **cannot reach the destination L_p** via the given routes.

(In a full system, an alternative route planner would re-route via a different path if one existed.)

2024 — Q4: ER + Perfect Square FOL (9 marks)

Question — 2024 Q4

- (a) In ER (Evidential Reasoning), how is a high-level attribute assessed through associated lower-level attributes? (1 mark)
- (b) What is the combined degree of belief in the original ER approach? (1 mark)
- (c) Consider the following sentences: (7 marks)
- (i) If a perfect square is divisible by a prime P , then it is also divisible by square of P
 - (ii) Every perfect square is divisible by some prime
 - (iii) 36 is a perfect square
 - (iv) Does there exist a prime q such that square of q divides 36?
- A) Translate the sentences above into FOL sentences. (2 marks)
- B) Prove (iv) by using Universal Elimination, Existential Elimination, Modus Ponens, and Existential Introduction. (1.5 marks)
- C) Convert the FOL sentences into Clausal Form by using Canonical Form (CF). (1.5 marks)
- D) Apply the Resolution method to prove (iv). (2 marks)

Part (a): Evidential Reasoning — Assessment of High-Level Attributes (1 mark)

Evidential Reasoning — 2024 Q4(a)

In **Evidential Reasoning (ER)**, a high-level attribute (e.g., “Disease D”) is assessed by combining probabilistic evidence from multiple lower-level attributes (e.g., symptoms, test results) using the Dempster–Shafer theory of belief functions. Each lower-level attribute provides a **basic probability assignment** (mass) distributed over hypotheses. The masses from all evidence sources are combined using **Dempster’s combination rule** (orthogonal sum), yielding a **belief measure** $\text{Bel}(H)$ and a **plausibility measure** $\text{Pl}(H)$ for the high-level hypothesis H .

Part (b): Combined Degree of Belief (1 mark)

Combined Degree of Belief — 2024 Q4(b)

The **combined degree of belief** in the original ER approach uses **Dempster’s rule**:

$$m_{1 \oplus 2}(A) = \frac{1}{1 - K} \sum_{B \cap C = A} m_1(B) \cdot m_2(C)$$

where $K = \sum_{B \cap C = \emptyset} m_1(B) \cdot m_2(C)$ is the **conflict factor** (measure of contradictory evidence).

The term $\frac{1}{1-K}$ normalizes the combined belief to sum to 1.

Belief $\text{Bel}(H) = \sum_{A \subseteq H} m(A)$ represents total committed belief; **Plausibility** $\text{Pl}(H) = 1 - \text{Bel}(\neg H)$ represents the upper bound of belief.

Part (c-A): FOL Translation (2 marks)**FOL Translation — 2024 Q4(c-A)**

Identical to 2021 Q5(b-i). See Section 3.

$$F_1 : \forall x \forall p \text{ PerfSq}(x) \wedge \text{Prime}(p) \wedge \text{Divides}(p, x) \rightarrow \text{Divides}(\text{square}(p), x)$$

$$F_2 : \forall x \text{ PerfSq}(x) \rightarrow \exists p \text{ Prime}(p) \wedge \text{Divides}(p, x)$$

$$F_3 : \text{PerfSq}(36)$$

$$\text{Goal} : \exists q \text{ Prime}(q) \wedge \text{Divides}(\text{square}(q), 36)$$

Part (c-B): Proof Using Inference Rules (1.5 marks)**Inference Rule Proof — 2024 Q4(c-B)**

Prove: $\exists q \text{ Prime}(q) \wedge \text{Divides}(\text{square}(q), 36)$

1. $\text{PerfSq}(36)$ [F_3]
2. $\text{PerfSq}(36) \rightarrow \exists p \text{ Prime}(p) \wedge \text{Divides}(p, 36)$ [**UE** $x/36$ on F_2]
3. $\exists p \text{ Prime}(p) \wedge \text{Divides}(p, 36)$ [**Modus Ponens** : 1 + 2]
4. $\text{Prime}(p_0) \wedge \text{Divides}(p_0, 36)$ [**EE** (Skolem const p_0): from 3]
5. $\text{Prime}(p_0)$ and $\text{Divides}(p_0, 36)$ [**And-Elim** from 4]
6. $\text{PerfSq}(36) \wedge \text{Prime}(p_0) \wedge \text{Divides}(p_0, 36) \rightarrow \text{Divides}(\text{square}(p_0), 36)$
[**UE** $x/36, p/p_0$ on F_1]
7. $\text{Divides}(\text{square}(p_0), 36)$ [**MP**: 1, 5, 6]
8. $\text{Prime}(p_0) \wedge \text{Divides}(\text{square}(p_0), 36)$ [**And-Intro**: 5+7]
9. $\exists q \text{ Prime}(q) \wedge \text{Divides}(\text{square}(q), 36)$ [**EI** : from 8, generalise p_0 to q]

Goal proved. ✓

Part (c-C): Canonical Form Conversion (1.5 marks)**Clause Form — 2024 Q4(c-C)**

Identical to 2021 Q5(b-ii). See Section 3.

Clause	Source
$C_1 : \{\neg \text{PerfSq}(x), \neg \text{Prime}(p), \neg \text{Divides}(p, x), \text{Divides}(\text{square}(p), x)\}$	F_1 (Steps 2,7)
$C_{2a} : \{\neg \text{PerfSq}(x), \text{Prime}(g(x))\}$	F_2 (Skolem fn g)
$C_{2b} : \{\neg \text{PerfSq}(x), \text{Divides}(g(x), x)\}$	F_2 (Skolem fn g)
$C_3 : \{\text{PerfSq}(36)\}$	F_3
$C_4 : \{\neg \text{Prime}(q), \neg \text{Divides}(\text{square}(q), 36)\}$	Negated goal

Part (c-D): Resolution Proof (2 marks)**Resolution Proof — 2024 Q4(c-D)**

Identical to 2021 Q5(b-iii). See Section 3 for full 7-step derivation.

Summary:

1. $C_3 + C_{2a} \{x/36\} \rightarrow \text{Prime}(g(36))$
2. $C_3 + C_{2b} \{x/36\} \rightarrow \text{Divides}(g(36), 36)$
3. $C_3 + C_1 \{x/36\} \rightarrow \neg\text{Prime}(p) \vee \neg\text{Divides}(p, 36) \vee \text{Divides}(\text{square}(p), 36)$
4. $(1)+(3) \{p/g(36)\} \rightarrow \neg\text{Divides}(g(36), 36) \vee \text{Divides}(\text{square}(g(36)), 36)$
5. $(2)+(4) \rightarrow \text{Divides}(\text{square}(g(36)), 36)$
6. $(1)+C_4 \{q/g(36)\} \rightarrow \neg\text{Divides}(\text{square}(g(36)), 36)$
7. $(5)+(6) \rightarrow \boxed{\square}$ (**Contradiction $\Rightarrow$ Goal Proved!**) ✓

Quick Exam Summary — Exam Patterns for This Topic

Every Year Patterns — Memorise These

1. **Marcus/Pompeii** (2020, 2022): Always 4 parts — (a) WFF translation (b) backward reasoning (c) clause form (d) resolution. Now changes year each paper (2021 in 2020 paper, 2023 in 2022 paper).
2. **Perfect Square FOL** (2021, 2024): Always same 4 sentences. 2021 asks for UE/EE/MP/EI proof. 2024 adds ER questions. Proof is always: Skolem $g(36) \rightarrow$ derive Prime+Divides $\rightarrow$ contradict negated goal.
3. **FOL Translation: Robot/Drone/Car** (2023): Declare predicates FIRST, then encode each action rule as $\forall$ sentence with $\rightarrow$.
4. **KB Resolution (Wumpus)** (2021 Q4d): Use $R_4(\neg B_{1,1})$ + clause from biconditional of R_2 to immediately derive contradiction. 2-step proof.
5. **Satisfiability check** (2021 Q4c, 2022 A-4a): Simplify algebraically first (use Distributivity, De Morgan). Then check truth table.
6. **Clause form algorithm asked directly** (2022, 2016): Know all 9 steps. Pay attention to Skolemization rule (constant vs. function depends on outer $\forall$).